FACULTY OF COMPUTING
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كلية الحاسبات وتقنية المعلومات
جامعة الملك عبدالعزيز

FCIT
KAU

Lab 04 Predicates and Quantifiers

Objective

Solving exercises from the textbook in chapter 1.4-1.5 (partial from 1.6)

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:
1. Will be able to identify the truth values of the quantifiers and nested quantifiers, negate the quantifiers.
2. They will be able to solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

1- Divide students to groups and let them to solve the given example.
2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

1. Let $N(x)$ be the statement "x has visited North Dakota," where the domain consists of the students in your school. Express each of these quantifications in English.

   a) $\exists x N(x)$ **b)** $\forall x N(x)$ **c)** $\neg\exists x N(x)$ **d)** $\exists x \neg N(x)$ **e)** $\neg\forall x N(x)$ **f)** $\forall x \neg N(x)$
2. Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.

   a) $\forall x(C(x) \rightarrow F(x))$ **b)** $\forall x(C(x) \wedge F(x))$ **c)** $\exists x(C(x) \rightarrow F(x))$ **d)** $\exists x(C(x) \wedge F(x))$
3. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.
   a) Someone in your class can speak Hindi.
   b) Everyone in your class is friendly.
   c) There is a person in your class who was not born in California.
   d) A student in your class has been in a movie.
   e) No student in your class has taken a course in logic programming.
4. Determine the truth value of each of these statements if the domain consists of all real numbers.

   a) $\exists x(x^3 = -1)$ **b)** $\exists x(x^4 < x^2)$ **c)** $\forall x((-x)^2 = x^2)$ **d)** $\forall x(2x > x)$
5. Determine the truth value of each of these statements if the domain consists of all integers.

   a) $\forall n(n^2 \geq 0)$ **b)** $\exists n(n^2 = 2)$ **c)** $\forall n(n^2 \geq n)$ **d)** $\exists n(n^2 < 0)$

6. Let S(x) be the predicate "x is a student," F(x) the predicate "x is a faculty member," and A(x, y) the predicate "x has asked y a question," where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.
   a) Lois has asked Professor Michaels a question.
   b) Every student has asked Professor Gross a question.
   c) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.
   d) Some student has not asked any faculty member a question.
7. Determine the truth value of each of these statements if the domain for all variables consists of all integers.

   a) $\forall n \exists m(n^2 < m$ **b)** $\exists n \forall m(n < m^2)$ **c)** $\forall n \exists m(n + m = 0)$ **d)** $\exists n \forall m(nm = m)$
8. Express the negations of each of these statements so that all negation symbols immediately precede predicates.

   a) $\forall x \exists y P(x, y) \lor \forall x \exists y Q(x, y)$**b)** $\forall x \exists y(P(x, y) \land \exists z R(x, y, z))$**c)** $\forall x \exists y(P(x, y) \rightarrow Q(x, y))$
9. Express the negation of these propositions using quantifiers, and then express the negation in English.
   a) Some drivers do not obey the speed limit.**b)** All Swedish movies are serious.
   c) No one can keep a secret.**d)** There is someone in this class who does not have a good attitude.
10. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

    a) $\exists x \forall y(xy = y)$**b)** $\forall x \forall y(((x < 0) \land (y < 0)) \rightarrow (xy > 0))$

    c) $\exists x \exists y((x^2 > y) \land (x < y))$**d)** $\forall x \forall y \exists z(x + y = z$
11. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?If George does not have eight legs, then he is not a spider. George is a spider. $\therefore$ George has eight legs
12. What rule of inference is used in each of these arguments?
    a) If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
    b) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn
13. For each of these arguments determine whether the argument is correct or incorrect and explain why.
    a) All students in this class understand logic. Xavier is a student in this class. Therefore, Xavier understands logic.**b)** Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major.

c) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.**d)** Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.